Machine-Learning-Based Transmission Line Fault and Load Identification via Time-Domain Reflectometry

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***Abstract*—Time-domain reflectometry (TDR) locates faults on a cable from the echoes of a launched step, but threshold-based analysis struggles with weak reflections, lossy and dispersive lines, and noise. We study whether machine learning can recover more from a single step-TDR trace, on three tasks: classify the fault (7 classes), locate it, and identify the termination (6 classes, with R_L, L or C). Training data come from a frequency-domain lossy-line simulator with skin effect and dielectric loss, validated against bounce-diagram and transient theory. We compare a conventional rule-based detector, LightGBM on 94 engineered features, and a multi-task 1-D ResNet. On 10,000 held-out traces at 30 dB SNR, the ResNet reaches a fault macro-F_1 of 0.848 ± 0.007 (LightGBM 0.790, conventional 0.622), a location error of 6.1 m mean and 1.6 m median, and a load accuracy of 96.3% (93.4% and 80.2%). The learned models detect resistive faults with $|\Gamma|$ down to about 0.05, whereas the threshold detector needs 0.1–0.2. They remain ahead from 60 to 15 dB and under shifts in line impedance, rise time, loss and cable length, but every method fails when the fault lies within 3 m of the load. On public laboratory records of an 80 m coaxial cable, echo timing locates a shield cutaway within 1.5 m of its documented position.**

***Index Terms*—Time-domain reflectometry, transmission lines, telegrapher's equations, skin effect, fault location, load identification, LightGBM, 1-D ResNet.**

I. Introduction

High-reliability electrical cabling systems form the critical nervous system of modern industrial infrastructure, ranging from aerospace avionics and naval power grids to high-speed data centers, telecommunication networks, and downhole energy wells. Transmission line degradation caused by thermal cycling, mechanical crushing, insulation weathering, chemical ingress, and localized conductor fracture inevitably introduces impedance discontinuities. If undetected, minor localized defects progressively escalate into catastrophic electrical failures, system outages, and hazardous short circuits.

Time-Domain Reflectometry (TDR) represents the primary non-destructive testing methodology for inspecting transmission lines [1], [2]. In traditional TDR, a high-speed voltage step or impulse is injected into the cable under test, and the reflected voltage waveform $v(0,t)$ is observed at the sending terminal. By exploiting the constant propagation velocity of transverse electromagnetic (TEM) waves along uniform transmission lines, the round-trip transit time of reflected echoes directly indicates the physical distance to the discontinuity, while the reflection amplitude and polarity characterize the local impedance mismatch.

Despite its conceptual elegance, practical TDR diagnosis suffers from severe physical non-idealities:

1) **High-Frequency Attenuation and Dispersion:** Real coaxial cables exhibit substantial frequency-dependent losses governed by conductor skin effect ($R(f) \propto \sqrt{f}$) and dielectric loss tangent ($G(f) \propto f$). These mechanisms severely smear and attenuate sharp incident wavefronts as propagation length increases, reducing the amplitude of distant fault echoes and obscuring reactive rise/fall signatures.
2) **Multiple Inter-Discontinuity Reflections:** When a fault discontinuity coexists with an imperfect terminal load, multiple reflections bounce back and forth within intermediate line segments. These secondary echoes overlap with genuine fault or load signatures, confounding classical single-echo assumptions.
3) **Environmental Noise and Masking Effects:** In field installations, ambient electromagnetic interference (EMI) degrades the signal-to-noise ratio (SNR), rendering weak resistive faults ($R_f \gg Z_0$ for shunt, $R_f \ll Z_0$ for series) indistinguishable from baseline thermal noise. Furthermore, severe discontinuities (such as a zero-ohm hard short or infinite-ohm open) reflect the entirety of the incident wave ($\Gamma = \pm 1$), completely masking downstream load terminations.

To overcome these physical bottlenecks, data-driven methods have been explored, starting with neural networks trained on simulated wiring-network reflectograms [3]. Convolutional neural networks (CNNs) have since been used to detect and characterise cable discontinuities [4], [5], while Gramian Angular Field (GAF) 2D representations [6] and compressed sensing TDR [7] have demonstrated superior robustness against background noise compared to manual thresholding. However, existing works predominantly frame the diagnosis as isolated single-task classification problems, restricting evaluations to

lossless or simplified lines, or omitting the simultaneous estimation of physical fault distance and complex reactive termination parameters (R_L, L, C).

In this work, we develop an end-to-end physics-informed machine learning framework that simultaneously solves three coupled diagnostic tasks from a single raw TDR trace:

- **Task 1 (T1) Fault Classification:** A 7-class taxonomy identifying pristine lines (`none`), hard discontinuities (`short`, `open`), soft reactive/resistive faults (`shunt_R`, `shunt_C`, `series_R`), and localized characteristic impedance variations (`z0_section`).
- **Task 2 (T2) Metric Fault Localization:** Precise continuous regression of the discontinuity location $d \in [0, l]$ reported in metres.
- **Task 3 (T3) Complex Load Characterization:** A 6-class classification (`open`, `short`, `matched`, `R`, `RL`, `RC`) combined with continuous parameter regression of termination resistance R_L and reactive time constants ($\tau = L/(R_L + Z_0)$ for a series RL load, $\tau = CR_LZ_0/(R_L + Z_0)$ for a parallel RC load).

We construct an exact, frequency-domain lossy transmission line cascade simulator based on the full telegrapher's equations and synthesize an 85,000-sample benchmark dataset spanning diverse cable lengths, loss scales, rise times, and SNR conditions. We evaluate and compare three distinct paradigms: (i) an optimized Conventional TDR rule-based solver, (ii) a Gradient Boosted Decision Tree (LightGBM [8]) utilizing 94 engineered physics features, and (iii) a 3-channel Deep 1D Residual Neural Network (1D-ResNet [9]) trained under a multi-task physics-masked loss function. Finally, we check the propagation physics on real hardware using public laboratory TDR records of an 80 m coaxial cable with a localised shield cutaway [10]: we estimate the velocity factor and locate the damage from echo timing. These are pulse-type records, so they test the physics rather than the step-trained models.

II. Electromagnetic Wave Theory & TDR Modeling

A. Telegrapher's Equations and Lossy Wave Propagation

TEM wave propagation along a uniform two-conductor transmission line is governed by the frequency-domain Telegrapher's Equations [1]:

$$\frac{dV(z,\omega)}{dz} = -Z(\omega)I(z,\omega), \quad \frac{dI(z,\omega)}{dz} = -Y(\omega)V(z,\omega) \tag{1}$$

where $Z(\omega) = R(\omega) + j\omega L$ and $Y(\omega) = G(\omega) + j\omega C$ represent the series impedance and shunt admittance per unit length, respectively. Decoupling (1) yields the Helmholtz wave equations:

$$\frac{d^2V(z,\omega)}{dz^2} - \gamma^2(\omega)V(z,\omega) = 0 \tag{2}$$

whose solutions comprise forward- and backward-traveling waves with complex propagation constant $\gamma(\omega) = \alpha(\omega) + j\beta(\omega) = \sqrt{Z(\omega)Y(\omega)}$ and characteristic impedance $Z_0(\omega) = \sqrt{Z(\omega)/Y(\omega)}$:

$$V(z,\omega) = V_0^+ e^{-\gamma(\omega)z} + V_0^- e^{+\gamma(\omega)z} \tag{3}$$

B. Lossy Coaxial Cable Physics (Skin Effect & Dielectric Dissipation)

To realistically model RG-58-like coaxial cables, we incorporate both conductor and dielectric losses:

1) **Skin Effect:** At radio frequencies, current concentrates within a surface layer of depth $\delta_s = \sqrt{2/(\omega\mu\sigma)}$. Consequently, the series resistance increases with the square root of frequency:
$$R(\omega) = R_{\text{dc}} + k_{\text{skin}}\sqrt{\frac{\omega}{2\pi}} \tag{4}$$
where $R_{\text{dc}} = 0.04\ \Omega/\text{m}$ is the DC resistance and $k_{\text{skin}} = 1.5\times10^{-4}\ \Omega/(\text{m}\cdot\sqrt{\text{Hz}})$ governs the skin depth boundary layer.
2) **Dielectric Loss:** The insulating polyethylene dielectric possesses a non-zero loss tangent $\tan\delta$, yielding a shunt conductance directly proportional to frequency:
$$G(\omega) = \omega C \tan\delta \tag{5}$$
with $\tan\delta = 2.0 \times 10^{-4}$.

The nominal phase velocity in the TEM mode is $u = v_f c = 1/\sqrt{LC}$, where $v_f \approx 0.66$ denotes the cable velocity factor and $c = 2.9979 \times 10^8$ m/s.

C. Reflection Coefficients and Discontinuity Boundary Conditions

When a propagating voltage wave encounters a localized impedance boundary Z_{load} at position z, the continuity of voltage and current dictates a reflected wave with voltage reflection coefficient Γ:

$$\Gamma = \frac{Z_{\text{load}} - Z_0}{Z_{\text{load}} + Z_0} \tag{6}$$

1) Hard Faults:

- **Short Circuit ($Z_f \to 0$):** $\Gamma = -1$. The reflected pulse has equal amplitude and reversed polarity, forcing total terminal voltage to zero upon arrival.
- **Open Circuit ($Z_f \to \infty$):** $\Gamma = +1$. The reflected pulse has equal amplitude and identical polarity, doubling the terminal voltage upon arrival.

2) Soft Reactive and Resistive Faults:

- **Shunt Faults (Z_f in parallel with line):** The effective impedance seen at the discontinuity is the parallel combination of Z_f and the downstream line impedance Z_0:
$$Z_{\text{eff}} = \frac{Z_f Z_0}{Z_f + Z_0} \implies \Gamma_{\text{shunt}} = \frac{-Z_0}{2Z_f + Z_0} \tag{7}$$
For a shunt resistor R_f, $\Gamma_{\text{shunt}} \in (-1, 0)$ produces a negative step echo. For a shunt capacitor C_f, the uncharged capacitor acts as an instantaneous short ($\Gamma \to -1$), recovering exponentially to $\Gamma \to 0$ as it charges, yielding a characteristic transient negative spike.
- **Series Faults (R_f in series with inner conductor):** The downstream impedance is $Z_0 + R_f$:
$$\Gamma_{\text{series}} = \frac{(Z_0 + R_f) - Z_0}{(Z_0 + R_f) + Z_0} = \frac{R_f}{2Z_0 + R_f} \tag{8}$$

Here $\Gamma_{\text{series}} \in (0, 1)$, generating a positive step echo.

- **Characteristic Impedance Section ($z_0_section$):** A finite segment of length w with altered geometry has local impedance Z_0'. Reflections occur at both the entry ($z = d$) with $\Gamma_1 = (Z_0' - Z_0)/(Z_0' + Z_0)$ and exit ($z = d+w$) with $\Gamma_2 = -\Gamma_1$, creating a rectangular pulse or flat pedestal of duration $\Delta t = 2w/u$.

D. Complex Load Terminations and Reactive Time Constants

At the terminal end of the line ($z = l$), we evaluate six load classes:

1) **Matched Load ($Z_L = Z_0$):** $\Gamma_L = 0$; no reflection returns from the load.
2) **Open and Short ($Z_L \to \infty$, $Z_L = 0$):** $\Gamma_L = +1$ and -1.
3) **Resistive Load ($Z_L = R_L$):** $\Gamma_L = (R_L - Z_0)/(R_L + Z_0)$.
4) **Inductive-Resistive Load (RL Series):** The load impedance $Z_L(s) = R_L + sL$ yields the reflection coefficient in the Laplace domain:

$$\Gamma_L(s) = \frac{sL + R_L - Z_0}{sL + R_L + Z_0} = 1 - \frac{2Z_0/L}{s + (R_L + Z_0)/L} \quad (9)$$

In response to an incident step V_{inc}, the reflected waveform exhibits an initial open-circuit-like step ($\Gamma \to +1$) decaying exponentially toward the resistive steady-state $\Gamma_\infty = (R_L - Z_0)/(R_L + Z_0)$ with time constant:

$$\tau_{\text{RL}} = \frac{L}{R_L + Z_0} \quad (10)$$

5) **Capacitive-Resistive Load (RC Parallel):** A resistor in parallel with a capacitor, $Z_L(s) = R_L/(1 + sR_LC)$, gives

$$\Gamma_L(s) = \frac{(R_L - Z_0) - sCR_LZ_0}{(R_L + Z_0) + sCR_LZ_0} \quad (11)$$

At the instant the step arrives the uncharged capacitor acts as a short ($\Gamma_0 = -1$), so the trace dips; as it charges the reflection relaxes to the resistive value $\Gamma_\infty = (R_L - Z_0)/(R_L + Z_0)$ with the time constant of C discharging through $R_L \parallel Z_0$:

$$\tau_{\text{RC}} = C\,\frac{R_LZ_0}{R_L + Z_0} \quad (12)$$

E. Lattice Bounce Diagram & Physics Masks

In a finite line of length l fed by a generator $V_g(t)$ with source resistance R_g, the initial incident wave launched into the line at $t = 0$ is:

$$V_{\text{inc}} = V_g \cdot \frac{Z_0}{Z_0 + R_g} \quad (13)$$

The wave propagates toward the fault at d and load at l, undergoing successive partial reflections and transmissions at the discontinuity, load (Γ_L), and generator ($\Gamma_g = (R_g - Z_0)/(R_g + Z_0)$). The round-trip transit time is defined as:

$$T_{\text{nom}} = \frac{2l}{u} = \frac{2l}{v_f c} \quad (14)$$

A fault at distance d produces its first echo at $t_f = 2d/u = (d/l)T_{\text{nom}}$, while the load echo arrives at $t_L = T_{\text{nom}}$.

Physical Masking Constraints:

- *Fault Location Masking:* When a cable is pristine (`none`), no fault discontinuity exists; hence the fault location d/l is physically undefined and must be masked out during both model training and metric computation.
- *Load Masking by Hard Discontinuities:* If the cable contains an ideal hard short ($\Gamma = -1$) or hard open ($\Gamma = +1$), the discontinuity acts as a perfect electromagnetic reflector. No forward wave energy penetrates past $z = d$ to interrogate the downstream cable or termination. Consequently, downstream load class and parameter predictions are unobservable and must be masked out for samples with `short` or `open` faults.

III. Simulation Engine & Dataset Generation

A. Exact Frequency-Domain Cascade Simulator

To avoid numerical dispersion artifacts inherent in finite-difference time-domain (FDTD) grid discretization, we implement an exact analytical frequency-domain cascade simulator [1]. A transmission line containing a localized discontinuity at $z = d$ and terminal load at $z = l$ is partitioned into two homogeneous transmission line sections separated by a two-port discontinuity matrix.

For a uniform section of length x with characteristic impedance $Z_0(\omega)$ and complex propagation constant $\gamma(\omega)$, the input impedance looking toward a load $Z_L(\omega)$ is transformed via:

$$Z_{\text{in}}(\omega, x) = Z_0(\omega)\frac{Z_L(\omega) + Z_0(\omega)\tanh(\gamma(\omega)x)}{Z_0(\omega) + Z_L(\omega)\tanh(\gamma(\omega)x)} \quad (15)$$

The cascade computation proceeds backward from load to source:

1) Evaluate the load impedance $Z_L(\omega)$ across discrete frequencies $\omega_k = 2\pi f_k$.
2) Transform $Z_L(\omega)$ across the downstream segment of length $l_2 = l - (d + w)$ to obtain $Z_{\text{in},2}(\omega)$.
3) Incorporate the discontinuity impedance:
   - For shunt faults: $Z_{\text{disc}}(\omega) = Z_{\text{in},2}(\omega) \parallel Z_f(\omega)$.
   - For series faults: $Z_{\text{disc}}(\omega) = Z_{\text{in},2}(\omega) + R_f$.
   - For impedance sections: transform through section length w with $Z_0'(\omega)$.
4) Transform $Z_{\text{disc}}(\omega)$ across the upstream segment of length $l_1 = d$ to obtain the total line input impedance $Z_{\text{in},1}(\omega) = Z_{\text{in}}(0, \omega)$.
5) The total voltage at the input terminal is then:

$$V(0, \omega) = V_g(\omega) \cdot \frac{Z_{\text{in}}(0, \omega)}{Z_{\text{in}}(0, \omega) + R_g} \quad (16)$$

B. Time-Domain Synthesis via Discrete Fourier Inversion

The incident voltage step is generated with a smooth transition characterized by 10%–90% rise time t_r:

$$v_g(t) = \frac{V_0}{2}\left[1 + \text{erf}\left(\frac{t - t_{\text{launch}}}{\sqrt{2}\sigma_r}\right)\right] \quad (17)$$

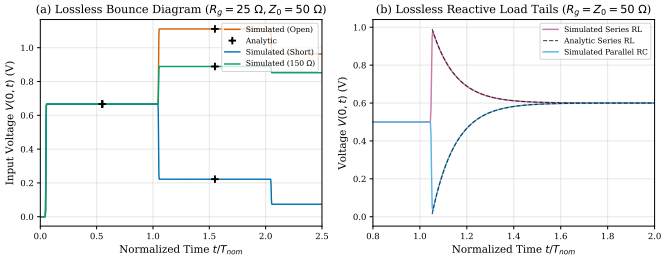


Fig. 1. Electromagnetic simulator validation: comparison against analytic lattice bounce diagrams on lossless lines ($R_g = 25\ \Omega$) and exact exponential transient tails for reactive RL and RC loads.

where $\sigma_r = t_r/2.563$. In the frequency domain, $V_g(\omega)$ is computed analytically, multiplied by the line transfer function (16), and synthesized in the time domain via the real inverse Fast Fourier Transform (IRFFT). We utilize $N_{\text{sim}} = 8192$ discrete frequency bins over a high-bandwidth spectrum to eliminate aliasing and DC wrap-around, resampling the final clean response to 1024 points uniformly spaced over $t/T_{\text{nom}} \in [0, 2.5)$ with the incident edge precisely aligned at $t/T_{\text{nom}} = 0.05$.

C. Simulator Validation

The simulator is checked against closed-form results before any data are generated (`1_validate_simulator.py`, Fig. 1). On a lossless line with $R_g = 25\ \Omega$ the first three bounce-diagram plateaus for open, short, 150 Ω and 10 Ω loads agree with the lattice-diagram values to within 3×10^{-4} (the small residual comes from the finite rise time). The fitted time constants of series RL and parallel RC loads agree with Eqs. (10) and (12) to a relative error below 10^{-6}, and the reflection coefficients measured for short and open faults equal -1 and $+1$ to within 10^{-6}. Fig. 2 shows one clean trace for each fault class.

D. Dataset Formulation & Out-of-Distribution Partitions

The dataset is generated by independently sampling physical cable, fault and load configurations (all ranges are in `config.yaml`; "log-uniform" means uniform in the logarithm):

- Cable length $l \in [20, 200]$ m; characteristic impedance $Z_0 \in [48, 52]\ \Omega$; velocity factor $v_f = 0.66\,\kappa$ with $\kappa \in [0.98, 1.02]$.
- Loss multiplier $s_{\text{loss}} \in [0.7, 1.3]$ applied to R_{dc} and k_{skin}; generator rise time $t_r \in [1, 5]$ ns; source resistance $R_g \in [47.5, 52.5]\ \Omega$.
- Faults, balanced over the 7 classes, at $d/l \in [0.10, 0.95]$: shunt $R_f \in [5, 2000]\ \Omega$ and series $R_f \in [1, 200]\ \Omega$ (log-uniform); shunt $C_f \in [20, 500]$ pF (log-uniform); impedance section $Z_0' \in [35, 45] \cup [55, 70]\ \Omega$ with width $w \in [0.5, 3]$ m.
- Loads, balanced over the 6 classes: matched $R_L \in [49, 51]\ \Omega$; $R_L \in [5, 500]\ \Omega$ (log-uniform, with $[40, 62]\ \Omega$ excluded for the resistive class so that it is distinguishable from matched); reactive time constants $\tau/T_{\text{nom}} \in [0.02, 0.3]$ (log-uniform).
- Measurement noise: white Gaussian with $\sigma = v_{\text{inc}} 10^{-\text{SNR}/20}$; during training the SNR is drawn uniformly in $[20, 50]$ dB for each trace.

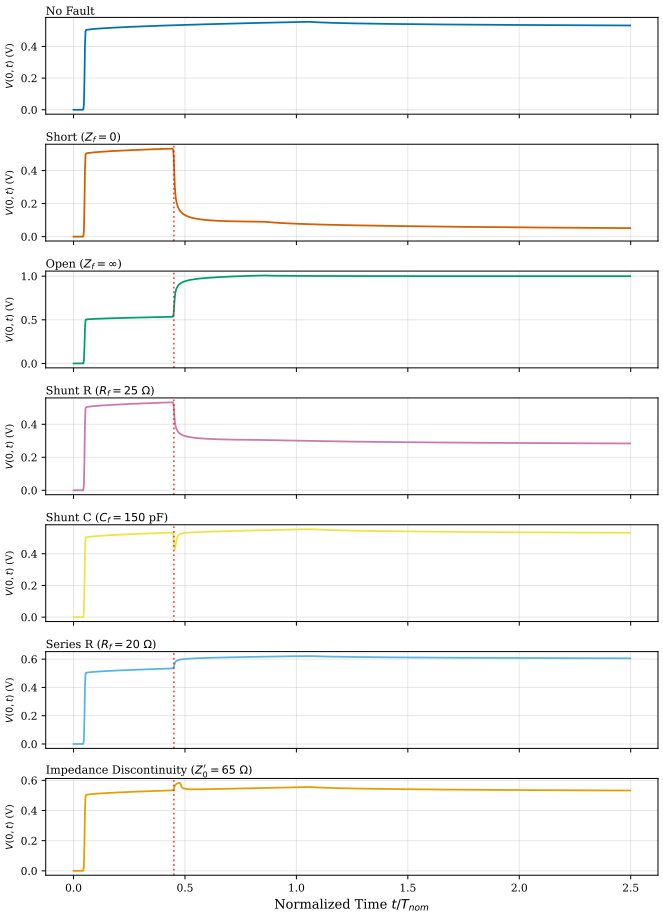


Fig. 2. Exemplar clean TDR waveforms across the 7 fault classes for an identical cable geometry ($l = 100$ m, $d = 40$ m, matched termination).

The production corpus comprises:

- **Training split:** 60,000 traces (seed 1).
- **Validation split:** 10,000 traces (seed 2), used for threshold tuning, early stopping, the learning curves and the ablations.
- **Test split:** 10,000 traces (seed 3), used for all headline results.
- **Out-of-distribution (OOD) subsets** ($5 \times 1{,}000$ traces, seed 4), each changing one factor beyond the training range:
  1) *Long cable:* $l \in [200, 300]$ m.
  2) *High loss:* $s_{\text{loss}} \in [1.5, 2.0]$.
  3) *Impedance shift:* $Z_0 \in [44, 47] \cup [53, 56]\ \Omega$.
  4) *Slow rise:* $t_r \in [5, 8]$ ns.
  5) *Near load:* fault 0.3–3 m from the load (so d/l can exceed the training maximum of 0.95); this subset

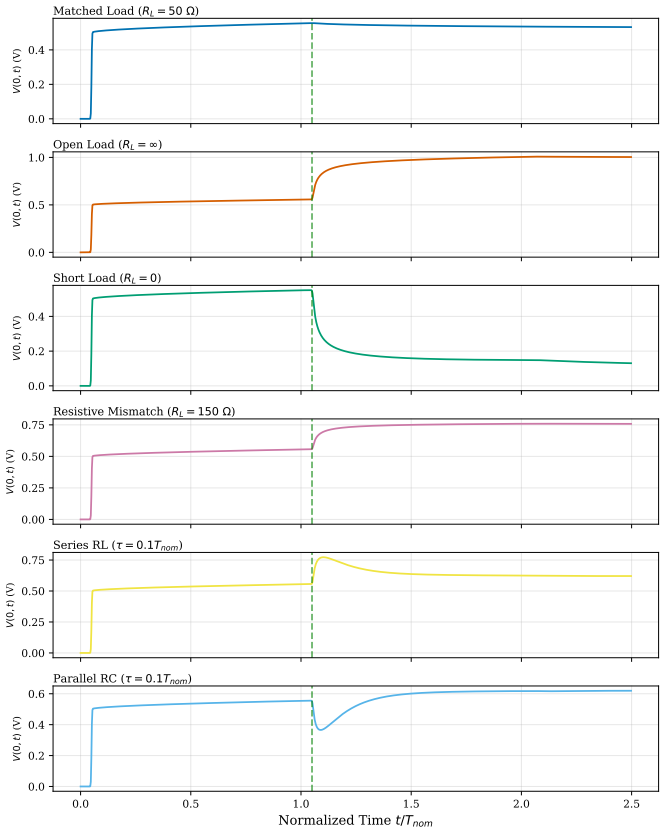


Fig. 3. Clean traces for the six load classes on a 100 m fault-free line. The dashed line marks the load echo; RL and RC loads produce exponential transients with time constant τ.

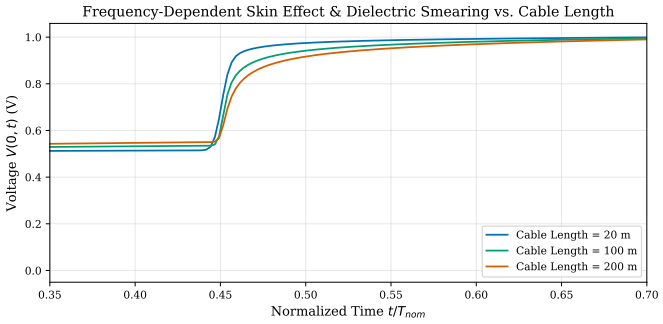


Fig. 4. Effect of conductor (skin) and dielectric loss: the open-load echo of a 20, 100 and 200 m cable, plotted on the normalised time axis. Longer cables give slower, more rounded edges.

contains no fault-free traces.

Generation is fully seeded and takes about 2 s per 1,000 traces (165 s for the full corpus). Fig. 3 shows the six load signatures, Fig. 4 the edge smearing caused by skin and dielectric loss as the cable gets longer, and Fig. 5 one trace at three SNR levels.

E. Laboratory Dataset: WISE-CASING

As a physical sanity check we use the public WISE-CASING laboratory TDR records [10] (Lawrence Berkeley National Laboratory): an 80 m RG-85 coaxial cable with shorted

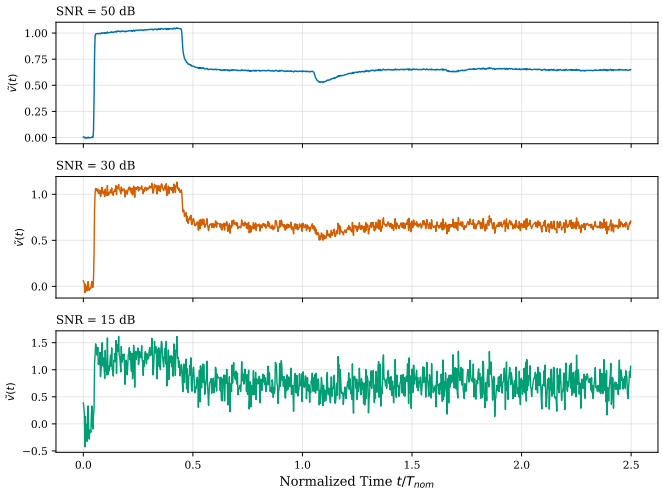


Fig. 5. The same trace (100 m line, 30 Ω shunt fault at 40 m, RC load) at 50, 30 and 15 dB SNR, after normalisation.

termination, measured undamaged and with a 120° shield cutaway at 40 m, plus records documented as 20, 10 and 5 m cables. These are *pulse-type* reflectometry records (the signal returns to zero between echoes), not step responses, so the step-trained models are not applied to them. Instead we verify the propagation physics on real hardware: the velocity factor is estimated from the undamaged 80 m record, and the damage is located from the ratio of echo times (Section VI-H).

IV. Diagnostic Methodologies

A. Unified Preprocessing Pipeline

Fig. 6 shows the complete pipeline. All three methods consume the same preprocessed inputs, produced by a single function. A raw trace $v(0, t)$ is normalized against the incident plateau step v_{inc}:

$$\tilde{v}(t) = \frac{v(0, t)}{v_{\text{inc}}} \tag{18}$$

Three parallel feature channels are constructed:

1) **Normalized Voltage $\tilde{v}(t)$:** Encodes baseline plateaus and reflection step levels.
2) **Numerical Derivative $d\tilde{v}/dt$:** Implemented via a Savitzky-Golay finite impulse response (FIR) filter (window length 11, polynomial degree 2, first derivative) wrapped in a 1D convolution layer. This sharpens inflection points and isolates high-frequency wave arrival edges while suppressing high-frequency AWGN.
3) **Dimensionless Coordinate Channel t/T_{nom}:** Linearly ramps from 0.0 to $2.5 \cdot (1023/1024)$, providing explicit temporal and spatial reference coordinates.

The resulting input tensor has shape $[3, 1024]$ across batch samples.

B. Conventional TDR Solver (Rule-Based Baseline)

The conventional baseline implements textbook TDR reasoning with no learned parameters other than a handful of

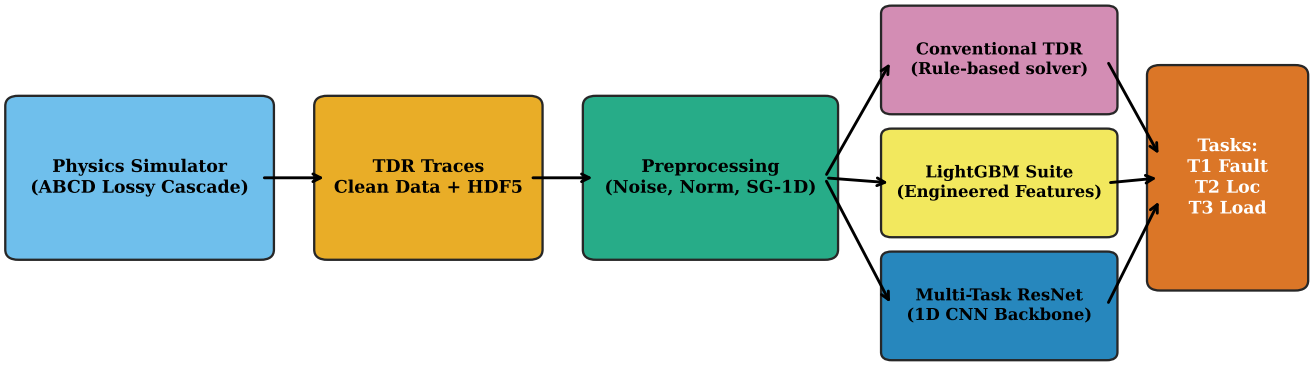


Fig. 6. End-to-End TDR Diagnostic Pipeline: from exact frequency-domain transmission line cascade simulation to 3-channel standardized preprocessing, feature engineering, and multi-model inference across Conventional, LightGBM, and Multi-Task ResNet1D architectures.

thresholds, all selected by grid search on the validation split (never on test) to maximise fault macro-F1 plus load macro-F1.

1) **Noise estimation:** $\hat{\sigma}_v = 1.4826\,\mathrm{MAD}(\Delta\tilde{v})/\sqrt{2}$ over $t/T \in [0.11, 2.45]$ (robust to the few edges present), and $\hat{\sigma}_{dv}$ as the MAD of the derivative channel.
2) **Edge detection:** peaks of $|d\tilde{v}|$ above $k_\sigma \hat{\sigma}_{dv}$ in the fault window $t/T \in [0.11, 1.00]$. A peak at sample p is accepted if its step $\hat{\Gamma} = \mathrm{med}\,\tilde{v}_s[p{+}3{:}p{+}13] - \mathrm{med}\,\tilde{v}_s[p{-}13{:}p{-}3]$ satisfies $|\hat{\Gamma}| \geq \max(0.02, c\,\hat{\sigma}_v)$. The first accepted edge is the fault candidate; none gives `none`.
3) **Hard faults:** decided only from the level after the edge relative to the level before it, $\rho = L_{\mathrm{after}}/L_{\mathrm{before}}$ with L_{after} the median over $[t_e{+}0.05, \min(t_e{+}0.25, 1)]$: $\rho < a \Rightarrow$ `short`, $\rho > b \Rightarrow$ `open`. The end-of-trace level is deliberately not used, because it is set by the load.
4) **Soft faults by shape:** a persistent step down/up gives `shunt_R/series_R`; an excursion that returns to the pre-edge level within the window of a short section is a transient: a positive bump, or a dip with comparable falling and rising edges, gives `z0_section`, and a sharp dip with slow recovery gives `shunt_C`.
5) **Location:** $\hat{d}/l = t_e/T - 0.05$, with t_e the half-amplitude crossing of the edge.
6) **Load (DC de-embedding):** from the settled end level L_{end} (median over $t/T \in [2.30, 2.45]$) the input impedance is $Z_{\mathrm{in}} = Z_0 L_{\mathrm{end}}/(2 - L_{\mathrm{end}})$. The estimated soft fault is removed (series: $Z_{\mathrm{in}} - R_f$; shunt: $(Z_{\mathrm{in}}^{-1} - R_f^{-1})^{-1}$, with R_f from $\hat{\Gamma}$), then the line resistance $r_{dc}l$ is subtracted to give $\hat{R}_L$. The over/undershoot after the load echo distinguishes `RL/RC`; $|\Gamma_{dc}|$ small gives `matched`; $\hat{R}_L$ thresholds give `open/short`; $\hat{\tau}/T$ comes from a log-linear fit of the tail.

C. LightGBM with 94 Physics Features

We engineer 94 features from the preprocessed 3-channel input:

- **Coarse waveform (64):** bin averages of $\tilde{v}$ over 64 equal time bins.
- **Level statistics (7):** final level, minimum, maximum and their times for $t/T > 0.11$, and mean levels over $[0.11, 1.0]$ and $[1.0, 2.5]$.
- **Edges (16):** for the four largest $|d\tilde{v}|$ peaks: time, signed peak height, local step size and half-maximum width.
- **First discontinuity (2):** first-edge reflection estimate $\hat{\Gamma}_1$ and impedance ratio $(1 + \hat{\Gamma}_1)/(1 - \hat{\Gamma}_1)$.
- **Load region (4):** level over $[0.90, 0.97]$, overshoot and undershoot relative to the final level, and $\hat{\tau}/T$ from a log-linear tail fit.
- **Noise (1):** robust noise estimate $\hat{\sigma}$.

Four LightGBM [8] models are trained: fault class (multiclass), location (regression on faulted samples), load class (multiclass where the load is observable) and load values ($\log_{10} R_L$ and $\log_{10} \tau/T$, each on its own mask). Each uses up to 3000 trees (learning rate 0.05, 63 leaves) with early stopping after 100 rounds on the validation set. Training uses two independent noisy copies of the training set (SNR drawn uniformly in [20, 50] dB per sample).

D. Deep Multi-Task 1D-ResNet Architecture

To exploit continuous temporal hierarchies and avoid manual feature design, we design a deep Multi-Task 1D-ResNet [9] operating directly on the $[3, 1024]$ input tensor:

- **Stem Layer:** A 1D convolution with kernel size 7, stride 2, padding 3 mapping $3 \to 32$ channels, followed by 1D Batch Normalization and ReLU activation, reducing sequence length to 512.

- **Four Residual Stages:** Each stage comprises two residual blocks with skip connections:
  - Stage 1: 32 channels, stride 1 (length 512).
  - Stage 2: 64 channels, stride 2 (length 256).
  - Stage 3: 128 channels, stride 2 (length 128).
  - Stage 4: 128 channels, stride 2 (length 64).
- **Latent Embedding:** the 128×64 feature map is flattened (8192 values; flattening, unlike global pooling, keeps the absolute time position needed for fault location), followed by 20% dropout, a linear projection $8192 \to 256$ and ReLU.
- **Task-Specific Prediction Heads:**
  - Head 1 (T1 Fault): Linear($256 \to 7$) producing class logits $\mathbf{z}_{\text{fault}}$.
  - Head 2 (T2 Location): Linear($256 \to 1$) followed by Sigmoid activation, constraining $\hat{d}/l \in [0, 1]$.
  - Head 3 (T3 Load Class): Linear($256 \to 6$) producing class logits $\mathbf{z}_{\text{load}}$.
  - Head 4 (T3 Load Values): Linear($256 \to 2$) regressing log-scaled parameters $[\log_{10} \hat{R}_L, \log_{10}(\hat{\tau}/T)]$.

E. Training Protocol

The ResNet is trained with AdamW (learning rate 2×10^{-3}, weight decay 10^{-4}, batch size 256) for at most 40 epochs, with a one-epoch linear warm-up followed by cosine decay, early stopping on the validation loss (patience 8), and mixed precision on GPU. Fresh measurement noise (SNR uniform in [20, 50] dB per sample) is drawn for every batch, so the network never sees the same noisy trace twice; noise injection and preprocessing run on the GPU. The validation set uses one fixed noisy copy at 30 dB. Channels are standardised with statistics from the training set. Main results are reported as mean $\pm$ standard deviation over three seeds.

F. Physics-Masked Multi-Task Loss Formulation

Joint multi-task optimization is governed by our physics-masked loss function:

$$\mathcal{L}_{\text{total}} = w_1 \mathcal{L}_{\text{fault}} + w_2 \mathcal{L}_{\text{loc}} + w_3 \mathcal{L}_{\text{load_cls}} + w_4 \mathcal{L}_{\text{load_reg}} \quad (19)$$

where loss balancing weights are set to $w_1 = 1.0, w_2 = 5.0, w_3 = 1.0, w_4 = 2.0$.

The component losses strictly respect the physical masking conditions:

1) **Fault Classification Loss (Unmasked):**

$$\mathcal{L}_{\text{fault}} = -\log\left(\frac{\exp(z_{\text{fault},y_1})}{\sum_{c=1}^{7} \exp(z_{\text{fault},c})}\right) \quad (20)$$

2) **Fault Location Loss (Masked on Fault Presence):** Evaluated only when true fault class is not `none` ($m_{\text{loc}} = \mathbb{I}(y_1 \neq \texttt{none})$):

$$\mathcal{L}_{\text{loc}} = \frac{\sum_{i=1}^{B} m_{\text{loc},i} \cdot \text{Smooth}_{L_1}(\hat{d}_i/l - d_i/l)}{\sum_{i=1}^{B} m_{\text{loc},i} + \epsilon} \quad (21)$$

3) **Load Classification Loss (Masked on Transmissibility):** Evaluated only when the fault is not a hard short or open ($m_{\text{load}} = \mathbb{I}(y_1 \notin \{\texttt{short}, \texttt{open}\})$):

$$\mathcal{L}_{\text{load_cls}} = \frac{\sum_{i=1}^{B} m_{\text{load},i} \cdot \text{CE}(\mathbf{z}_{\text{load},i}, y_{3,i})}{\sum_{i=1}^{B} m_{\text{load},i} + \epsilon} \quad (22)$$

4) **Load Value Regression Loss (Masked):**

$$\mathcal{L}_{\text{load_reg}} = \frac{\sum_{i=1}^{B} m_{\text{load},i} \cdot \text{Smooth}_{L_1}(\hat{\mathbf{r}}_i - \mathbf{r}_i)}{\sum_{i=1}^{B} m_{\text{load},i} + \epsilon} \quad (23)$$

Masking guarantees that non-existent targets never inject corrupting gradients into the shared residual backbone.

V. Experimental Protocol & Evaluation Metrics

A. Evaluation Metrics

Diagnostic performance across the three tasks is measured using the following standardized metrics:

1) **Task 1 (Fault Classification):**
   - Overall Classification Accuracy (%).
   - Macro-Averaged F_1 Score: unweighted mean across all 7 classes, penalizing poor performance on minority fault categories.
   - 7×7 Confusion Matrices.
2) **Task 2 (Fault Localization):** computed over all samples whose true class is not `none`, whatever class was predicted, so a missed fault still counts with the location the method outputs (the conventional detector outputs $d/l = 0.5$ when it finds no edge):
   - Mean Absolute Error in metres: $\text{MAE} = \frac{1}{N_{\text{fault}}} \sum_{i=1}^{N_{\text{fault}}} |\hat{d}_i - d_i|$.
   - Median Absolute Error in metres: robust to extreme outliers.
   - 90th Percentile Error (P_{90} in metres): captures boundary failure cases.
   - Sub-Metre Localization Rate: fraction of faulty samples localized within 1.0 m of ground truth ($\% < 1.0$ m).
3) **Task 3 (Load Characterization):** Computed strictly over samples where the load is physically observable ($y_{\text{fault}} \notin \{\texttt{short}, \texttt{open}\}$):
   - Load Classification Accuracy (%) and Macro-F_1 across 6 classes.
   - Mean Absolute Percentage Error (MAPE) on termination resistance: $\text{MAPE}(R_L) = \frac{100\%}{N} \sum |\hat{R}_{L,i} - R_{L,i}|/R_{L,i}$.
   - MAPE on derived inductance $\hat{L} = \hat{\tau}(\hat{R}_L + Z_0)$ for true RL loads.
   - MAPE on derived capacitance $\hat{C} = \hat{\tau}/(\hat{R}_L + Z_0)$ for true RC loads.

B. Experimental Suite (E1–E8)

Our experimental protocol comprises eight systematic investigations:

- **E1: Benchmark Comparison:** Conventional TDR, LightGBM and 1D-ResNet on the held-out test split at 30 dB,

with one fixed noise realisation (seed 1234). The ML models are trained with three seeds and reported as mean $\pm$ standard deviation.

- **E2: Noise Sweep:** the test split at seven SNR levels $\{60, 50, 40, 30, 25, 20, 15\}$ dB with the seed-0 models and one fixed noise realisation per level. 15 and 60 dB lie outside the 20–50 dB training range.
- **E3: Out-of-Distribution Generalization:** the five OOD subsets of Section III (long cable 200–300 m, loss $\times 1.5$–2, Z_0 shifted to 44–47 or 53–56 Ω, rise time 5–8 ns, fault 0.3–3 m from the load), at 30 dB, with the seed-0 models.
- **E4: Error Stratification:** location MAE of the seed-0 ResNet as a function of relative position d/l and of the nominal reflection strength $|\Gamma|$ of the fault, and the detection recall (predicted class $\neq$ `none`) of every method per $|\Gamma|$ bin for the resistive and Z_0-section faults.
- **E5: Sample Efficiency & Learning Curves:** LightGBM and ResNet (seed 0) trained on fractions $\{0.05, 0.10, 0.25, 0.50, 1.0\}$ of the training split and scored on the *validation* split.
- **E6: Ablation Studies** (seed 0, scored on the *validation* split):
  1) Input representation: 3-channel $(\tilde{v}, d\tilde{v}/dt, t/T)$ vs. 2-channel $(\tilde{v}, d\tilde{v}/dt)$ vs. single-channel $(\tilde{v})$.
  2) Multi-task synergy: Joint multi-task learning vs. three independent single-task networks dedicated to T1, T2, and T3.
- **E7: Explainability via SHAP Analysis:** mean absolute SHAP values (TreeExplainer [11]) of the seed-0 LightGBM fault classifier on 2,000 test traces at 30 dB, averaged over the seven classes.
- **E8: Physical Consistency on Laboratory TDR Data:** on the WISE-CASING pulse-TDR records [10], measure round-trip echo times, estimate the velocity factor from the undamaged 80 m cable, locate the 40 m cutaway by echo-time ratio, and flag records whose echo times contradict their documented length.

VI. Experimental Results & Evaluation

All results come from the full run on a Kaggle T4 GPU (60,000 training traces). The complete pipeline, from data generation to figures, took 135 min: 3 min to generate the data, 4 min to tune the conventional detector, 63 min for the seven LightGBM suites, 51 min for the twelve ResNet runs (about 9.4 s per epoch) and 14 min for evaluation.

A. E1: Main Comparison on the Test Split

Table I gives the headline comparison at 30 dB. The multi-task ResNet is the best method for:

- fault classification: macro-F_1 0.848 ± 0.007, against 0.790 for LightGBM and 0.622 for the conventional detector;
- fault location, both in the mean (MAE 6.06 ± 0.63 m) and in the tail ($P_{90} = 17.3$ m);
- load classification: $96.3\% \pm 0.8\%$.

LightGBM is second on all three tasks and has the lowest error on the load resistance (MAPE 12.1%). The ResNet has the lowest errors on the derived inductance and capacitance (18.6% and 17.2%, against 22.6% and 24.3% for LightGBM). The conventional detector's time-constant estimate is unreliable on lossy lines, so its L and C errors exceed 130%.

The location metrics reveal two different error profiles.

- *Conventional detector.* It has the best median error (0.91 m) and the highest share of faults located within 1 m (51.7%). When it finds the echo edge, it times it to the nearest output sample, which is $2.5\,l/2048 \approx 0.12$ m on a 100 m cable. However, it finds an edge for only 66.6% of the faulted traces. A missed fault is assigned $d/l = 0.5$, and these misses produce its 36 m P_{90}.
- *Learned models.* They detect far more faults, so their error tails are much shorter. Their regression output is less sharp, which gives a larger median error (1.4–1.6 m).

The ResNet's seed spread comes mostly from seed 2, which early stopping ended after 20 epochs (Section VII).

Fig. 7 shows where the classification errors occur. Both learned models recognise short and open faults almost perfectly (per-class $F_1 = 1.00$ for the ResNet). Nearly all of their remaining fault errors are weak faults predicted as `none`: for the seed-0 ResNet, 21% of shunt capacitances and 30% of Z_0 sections are called fault-free. In the other direction, 7% of fault-free lines are flagged as Z_0 sections. The conventional detector shows the same pattern much more strongly, reporting 49% of shunt-C and 66% of Z_0-section faults as `none`.

For the load, the learned models keep at least 88% recall on every class. Their errors are mostly between the R class and the reactive classes; for example, LightGBM labels 7% of RL loads as R. The conventional detector assigns 22–27% of the reactive loads and 16% of the matched loads to R.

B. E2: Noise Sweep

Table II and Fig. 8 show performance from 60 to 15 dB. The ordering ResNet $>$ LightGBM $>$ conventional holds at every SNR, for fault F_1, location MAE and load accuracy.

At high SNR the learned models approach the limits of the task. At 60 dB the ResNet reaches $F_1 = 0.986$, a location MAE of 1.21 m and a load accuracy of 99.2%. The conventional detector levels off at $F_1 \approx 0.74$ and about 81.5% load accuracy above 40 dB. Its remaining errors therefore come from its rules (weak and frequency-dependent echoes, lossy load edges) and not from noise.

Below the 20–50 dB training range the two learned models degrade differently. Between 20 and 15 dB, LightGBM's load accuracy falls from 86.4% to 63.0%, while the ResNet's falls only from 91.6% to 84.3%. LightGBM depends on hand-crafted edge features (peak times, step sizes), and these become unreliable once noise peaks are as large as the echoes.

C. E3: Out-of-Distribution Generalisation

Table III and Fig. 9 report the five OOD subsets. As an in-distribution reference, the seed-0 ResNet scores $F_1 = 0.854$, MAE 5.67 m and 96.9% load accuracy at 30 dB.

TABLE I

MAIN COMPARISON ON THE HELD-OUT TEST SPLIT (10,000 TRACES, 30 DB). LIGHTGBM AND RESNET1D: MEAN ± STANDARD DEVIATION OVER THREE TRAINING SEEDS. THE CONVENTIONAL DETECTOR IS DETERMINISTIC.

Model	Fault Acc (%)	Fault Macro-F1	Loc MAE (m)	Loc Median (m)	Loc P90 (m)	Loc <1m (%)	Load Acc (%)	Load Macro-F1	MAPE R (%)	MAPE L (%)	MAPE C (%)
Conventional TDR	61.83	0.6224	9.47	0.91	36.06	51.7	80.19	0.8123	38.2	134.3	140.5
LightGBM	78.27 ± 0.07	0.7902 ± 0.0009	7.67 ± 0.04	1.42 ± 0.02	25.66 ± 0.29	42.5 ± 0.2	93.40 ± 0.11	0.9348 ± 0.0011	12.1 ± 0.0	22.6 ± 0.3	24.3 ± 0.2
Multi-Task ResNet1D	84.38 ± 0.83	0.8483 ± 0.0073	6.06 ± 0.63	1.64 ± 0.56	17.30 ± 1.07	38.2 ± 8.0	96.34 ± 0.83	0.9637 ± 0.0082	16.9 ± 6.1	18.6 ± 4.9	17.2 ± 3.7

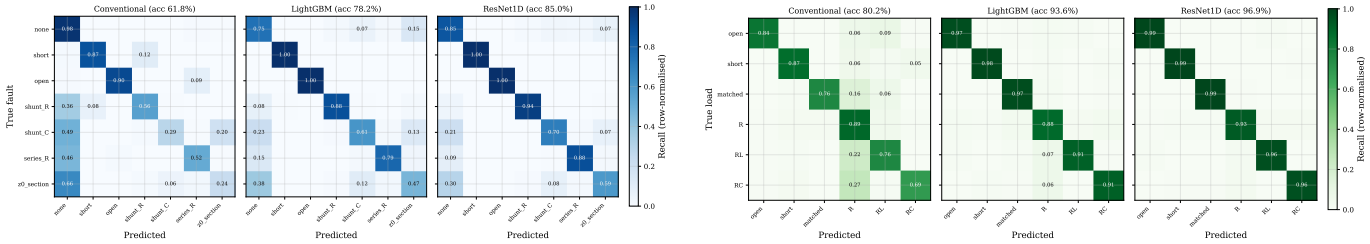


Fig. 7. Row-normalised confusion matrices on the test split at 30 dB (seed-0 models). Left: fault class (7 classes). Right: load class, over traces where the load is observable (no short or open fault).

TABLE II

NOISE SWEEP ON THE TEST SPLIT (SEED-0 MODELS). TRAINING USED SNR UNIFORM IN 20–50 DB, SO 15 AND 60 DB ARE OUTSIDE THE TRAINING RANGE.

SNR (dB)	Conventional F1	Conventional MAE (m)	Conventional load acc (%)	LightGBM F1	LightGBM MAE (m)	LightGBM load acc (%)	ResNet F1	ResNet MAE (m)	ResNet load acc (%)
60	0.741	5.36	81.5	0.932	2.78	96.2	0.986	1.21	99.2
50	0.739	5.30	81.6	0.922	3.11	96.5	0.982	1.22	99.1
40	0.728	5.75	81.9	0.882	4.42	95.8	0.953	2.10	98.7
30	0.622	9.47	80.2	0.789	7.66	93.6	0.854	5.67	96.9
25	0.547	11.45	68.5	0.726	9.77	90.6	0.785	7.97	94.9
20	0.469	13.17	59.3	0.662	12.08	86.4	0.719	10.26	91.6
15	0.379	15.54	55.6	0.566	15.31	63.0	0.621	13.48	84.3

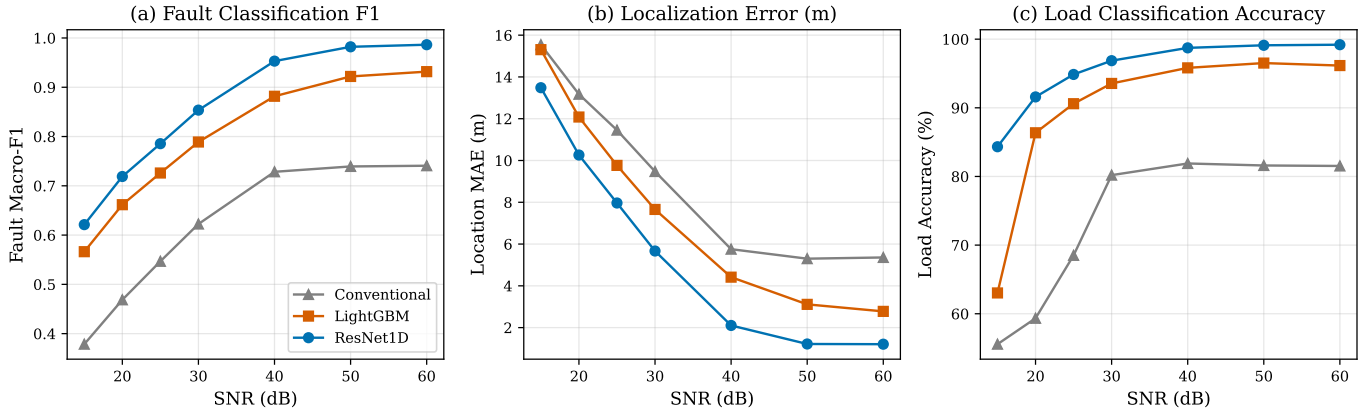


Fig. 8. Fault macro-F_1, location MAE and load accuracy against SNR for the three methods.

- *Impedance shift and slow rise.* Shifts of up to $\pm 6\ \Omega$ in Z_0 and rise times of 5–8 ns cost almost nothing (F_1 0.840 and 0.847).
- *High loss.* A 1.5–2× higher loss costs about 0.04 in F_1 and 5 points of load accuracy.
- *Long cable.* Cables of 200–300 m are harder: F_1 drops to 0.68 and the MAE rises to 22 m. The edges are smeared more strongly by loss, and a given error in d/l is worth more metres on a longer cable. Load classification still holds at 94%.
- *Near load.* With the fault 0.3–3 m before the load, all three methods fail ($F_1 \leq 0.28$, MAE ≥ 40 m). The fault and load echoes arrive only 3–30 ns apart. That is no more than the width of an edge after dispersion on a lossy line, so the two echoes merge into one. In addition, $d/l > 0.95$ never occurs in training.

On every subset the three methods keep the same ranking as in distribution.

D. E4: Where the Errors Come From

For the resistive and Z_0-section faults (4,332 test traces, the classes with a single nominal Γ), the ResNet's location error depends strongly on reflection strength (Fig. 10, bottom). For $|\Gamma| \geq 0.2$ the MAE is 1.0–1.7 m. For $|\Gamma| < 0.2$, which

TABLE III
OUT-OF-DISTRIBUTION SUBSETS (1,000 TRACES EACH, 30 DB, SEED-0 MODELS).

OOD subset	Conventional F1	Conventional MAE (m)	Conventional load acc (%)	LightGBM F1	LightGBM MAE (m)	LightGBM load acc (%)	ResNet F1	ResNet MAE (m)	ResNet load acc (%)
long_cable	0.469	27.85	79.7	0.574	24.58	85.5	0.682	22.19	94.0
high_loss	0.532	11.27	73.5	0.710	9.64	83.3	0.814	7.75	91.6
z0_shift	0.616	9.91	76.4	0.761	8.39	91.9	0.840	6.15	95.5
slow_rise	0.609	9.60	79.8	0.753	7.72	93.6	0.847	6.09	97.3
near_load	0.058	53.21	72.5	0.232	42.05	82.0	0.278	40.21	82.6

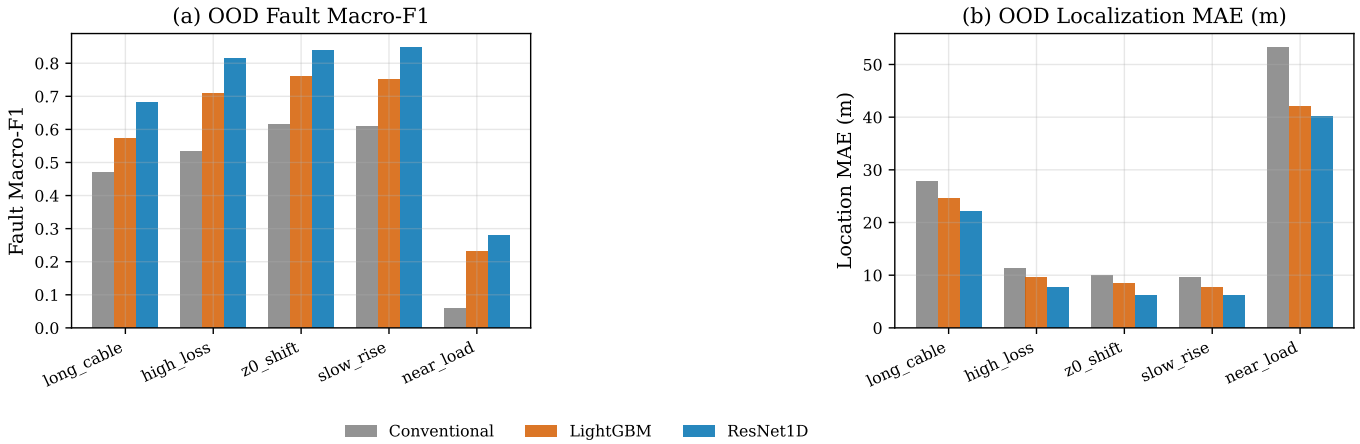


Fig. 9. Fault macro-F_1 (a) and location MAE (b) on the five OOD subsets.

covers 3,040 of these traces, it is 14.5 m. The location error is therefore concentrated in weak discontinuities.

Plotted against position over all faulted traces (Fig. 10, top), the error is U-shaped. It is lowest near the middle of the line (3.5 m at $d/l \approx 0.57$) and highest at the ends (10.1 m at $d/l \approx 0.14$ and 14.3 m at $d/l \approx 0.91$). This is what one expects from a regression head that falls back towards the centre of the line when it cannot see the echo. The same behaviour explains why the median error is much smaller than the mean.

Table IV measures the detection limit directly.

- *Conventional detector.* It finds fewer than a quarter of the resistive faults with $|\Gamma| < 0.1$, about 77% for $0.1 \le |\Gamma| < 0.2$ and 98% above 0.2. At 30 dB the per-sample noise is $\sigma \approx 0.032$ in normalised units, so this limit lies at three to six times σ, consistent with its $k_\sigma = 3$ threshold.
- *Learned models.* Both lower the detectable reflection by a factor of two or more. They detect 94–100% of the resistive faults with $0.05 \le |\Gamma| < 0.1$, and the ResNet still detects 72–79% of those below 0.05. A step that persists for the rest of the trace can be found by integrating over many samples, which a per-sample threshold cannot do.

Z_0 sections remain the hardest class for every method. Their echo is a positive–negative pair separated by the section's round-trip time ($2w/u \approx$ 5–30 ns). The two halves partly cancel once rise time and loss smear the edges together.

E. E5: Sample Efficiency

The learning curves (Table V and Fig. 11, validation split) show that the ResNet is also more data-efficient. Trained on 5% of the data (3,000 traces), it reaches $F_1 = 0.799$. This already exceeds LightGBM trained on all 60,000 traces (0.789).

Both curves are still rising at the full training set. For the ResNet, going from 50% to 100% of the data adds 0.007 in F_1 and lowers the MAE by 0.4 m. More simulated data would therefore still help, but only modestly.

TABLE IV
DETECTION RECALL VERSUS NOMINAL REFLECTION STRENGTH ON THE TEST SPLIT AT 30 DB (SEED-0 MODELS). EACH CELL GIVES DETECTED (PREDICTED $\neq$ NONE) / CORRECTLY CLASSIFIED, IN %. SHUNT-C REFLECTION DEPENDS ON FREQUENCY, SO IT IS POOLED.

Fault class	abs(Gamma) bin	n	Conventional	LightGBM	ResNet1D
shunt_R	[0.00, 0.05)	318	1 / 0	63 / 51	79 / 74
shunt_R	[0.05, 0.10)	191	24 / 23	99 / 97	100 / 100
shunt_R	[0.10, 0.20)	176	77 / 76	99 / 98	100 / 99
shunt_R	[0.20, 0.40)	246	98 / 98	100 / 100	100 / 100
shunt_R	[0.40, 1.00)	504	98 / 76	100 / 98	100 / 100
series_R	[0.00, 0.05)	465	4 / 3	57 / 39	72 / 64
series_R	[0.05, 0.10)	190	24 / 24	94 / 94	98 / 98
series_R	[0.10, 0.20)	219	78 / 76	100 / 100	100 / 100
series_R	[0.20, 0.40)	258	98 / 98	100 / 100	100 / 100
series_R	[0.40, 1.00)	284	98 / 90	100 / 100	100 / 100
shunt_C	all (frequency dependent)	1434	51 / 29	77 / 61	79 / 70
z0_section	[0.00, 0.05)	75	5 / 0	33 / 19	24 / 16
z0_section	[0.05, 0.10)	513	19 / 12	50 / 37	60 / 51
z0_section	[0.10, 0.20)	893	45 / 33	72 / 54	80 / 68

TABLE V
LEARNING CURVES (SEED 0, VALIDATION SPLIT, 30 DB).

Train fraction	LightGBM F1	LightGBM MAE (m)	ResNet F1	ResNet MAE (m)
0.05	0.7254	9.28	0.7990	8.74
0.1	0.7455	8.89	0.8229	8.10
0.25	0.7688	8.50	0.8393	6.87
0.5	0.7762	8.31	0.8513	6.10
1	0.7893	8.03	0.8578	5.71

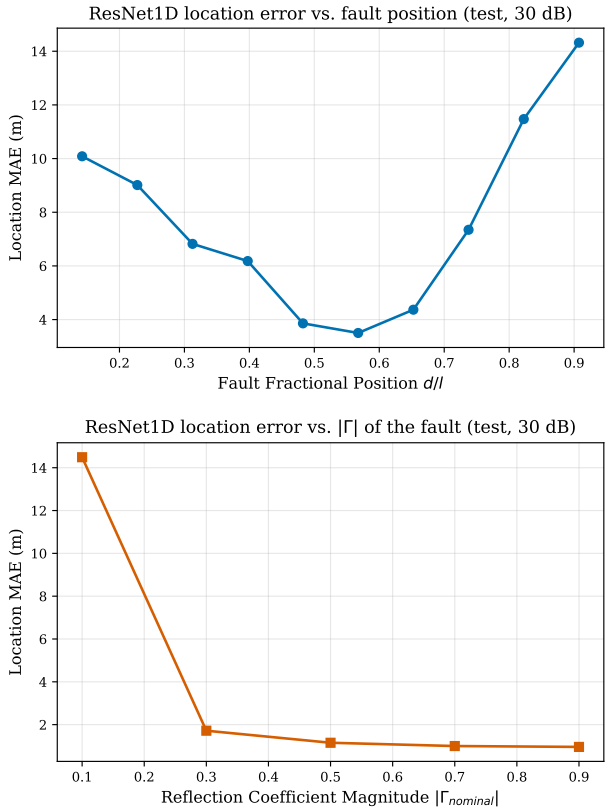


Fig. 10. ResNet location MAE on the test split (30 dB) by fault position d/l (top) and by nominal reflection strength $|\Gamma|$ of resistive and Z_0-section faults (bottom).

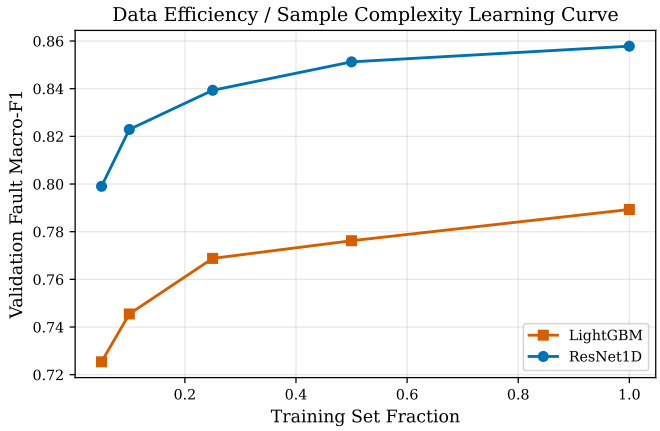


Fig. 11. Validation fault macro-F_1 against the fraction of the training split used.

F. E6: Ablations

Table VI and Fig. 12 (validation split, seed 0) answer two questions.

Are the derivative and coordinate channels needed? No. The voltage channel alone gives $F_1 = 0.854$, MAE 5.76 m and 97.1% load accuracy, within noise of the full three-channel model (0.858, 5.71 m, 97.2%). The first convolution can learn its own differentiating filter. The flattening head keeps absolute position, so the network does not need a coordinate channel to know *when* an echo occurs.

TABLE VI
RESNET ABLATIONS (SEED 0, VALIDATION SPLIT, 30 DB). "–" MARKS TASKS A SINGLE-TASK NETWORK WAS NOT TRAINED FOR. THE LAST COLUMN GIVES THE EPOCHS ACTUALLY RUN AND, IN BRACKETS, THE EPOCH OF THE SAVED CHECKPOINT.

Configuration	Fault macro-F1	Loc MAE (m)	Load acc (%)	Epochs run (best)
Full ResNet1D (v, dv, coord)	0.8578	5.71	97.22	40 (39)
No coord channel (v, dv)	0.8369	7.65	95.67	15 (7)
Voltage only (v)	0.8544	5.76	97.05	40 (39)
Single-task: fault	0.8536	–	–	40 (40)
Single-task: location	–	4.48	–	40 (40)
Single-task: load	–	–	97.02	40 (38)

The $(\tilde{v}, d\tilde{v}/dt)$ run scores lower (0.837, 7.65 m), but this does not show that the coordinate channel matters. Early stopping ended that run after 15 epochs, with its best checkpoint at epoch 7, whereas every other ablation trained for 40 epochs. The voltage-only model also lacks the coordinate channel and matches the full model. The shortfall is therefore best attributed to the shorter training, not to the missing channel.

Does multi-task learning help? For classification it is neutral: separate fault and load networks reach the same F_1 (0.854) and load accuracy (97.0%) as the shared network. For location it costs accuracy: a network trained only for location reaches an MAE of 4.48 m, against 5.71 m for the multi-task model. The shared backbone is a compromise between the three objectives, and the regression task loses most. In return, the multi-task model needs one network and one forward pass instead of three.

G. E7: What LightGBM Looks At

Fig. 13 ranks the inputs of the LightGBM fault classifier by mean |SHAP|. Two features stand out: the averaged trace levels over $t/T_{\mathrm{nom}} = 1.02$–$1.05$ and 0.98–1.02. Both lie just before the load echo, which arrives at $t/T_{\mathrm{nom}} \approx 1.05$.

These levels are informative for a simple reason. Before the load echo arrives, the normalised trace is close to 1 on a fault-free line and close to $1+\Gamma_f$ after a resistive or hard fault. A single level just before the load echo therefore summarises whether and how the line has changed. The next most important features are the times of the strongest edges (`edge1_time`, `edge2_time`), which carry the fault position. The explicit physics features (first-edge $\hat{\Gamma}$ and Z/Z_0) rank lower because the binned levels already contain the same information.

H. E8: Physical Consistency on Laboratory TDR Records

Table VII lists the round-trip time of the strongest echo in each WISE-CASING record [10], and Fig. 14 shows the records.

Velocity factor. The shorted, undamaged 80 m cable gives a negative echo after 0.836 μs, so $v_f = 2l/(ct) = 0.638$. This is 3% below the 0.66 assumed by our simulator, and we use it as the estimate for the other records.

Damage location. In the damaged record the strongest echo arrives after 0.402 μs. By echo-time ratio this places the 120°

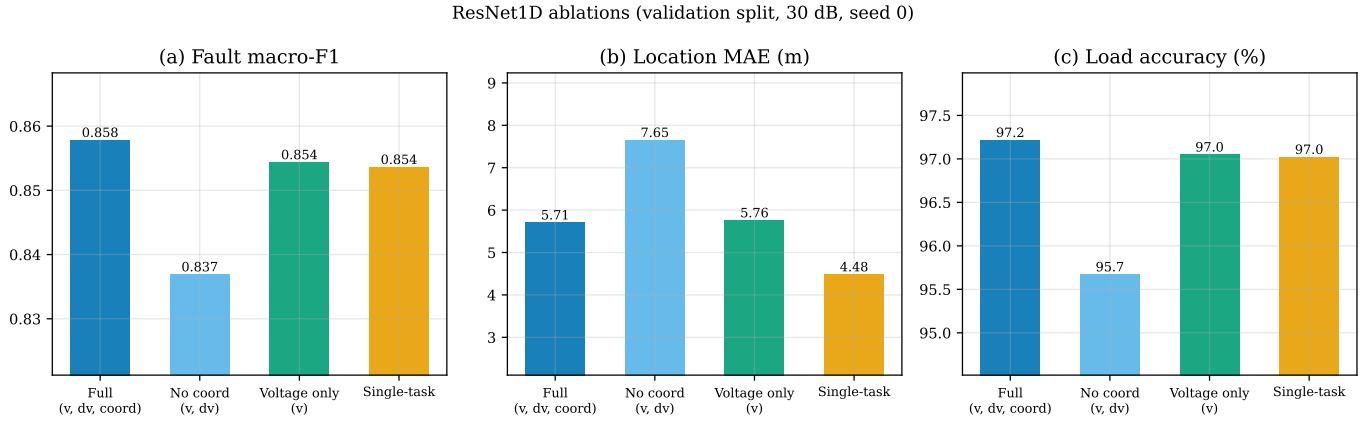


Fig. 12. ResNet ablations on the validation split. Each panel shows only the runs trained for that task.

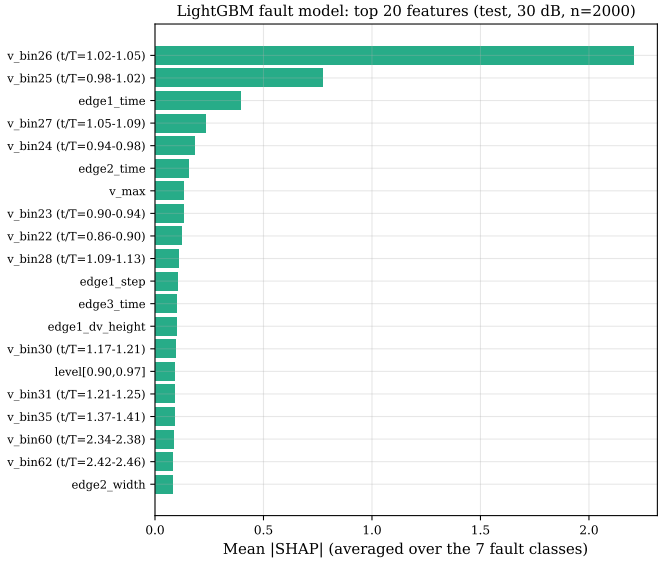


Fig. 13. Top 20 LightGBM fault-classifier features by mean |SHAP| (2,000 test traces, 30 dB, averaged over the seven classes). `v_binNN` is the mean of $\tilde{v}$ over the indicated t/T_{nom} interval; `edgeK` is the K-th strongest edge of $d\tilde{v}/dt$.

shield cutaway at 38.5 m, against the documented 40 m: an error of -1.5 m, or 1.9% of the cable length (Table VIII). At the nominal velocity factor the estimate would be 39.8 m. One place in the catalogue gives the cable length as 85 m instead of 80 m. With 85 m, the velocity factor would be 0.678 and the cutaway would lie at 40.9 m, so the localisation is within 1.5 m either way.

Echo polarity. The cutaway echo is *positive*, which means the line impedance rises locally. This is expected: removing part of the shield lowers the local capacitance and raises the local inductance. It is the physical counterpart of our Z_0-section class with $Z_0' > Z_0$. The damage echo is, however, larger than the recorded incident pulse, which a passive discontinuity cannot produce. We therefore use only its timing, not its amplitude.

Inconsistent records. The three records documented as 20, 10 and 5 m cables do not match their labels: their echoes correspond to 80.6, 38.5 and 13.1 m at the estimated velocity factor. The "20 m" echo time (0.842 μs) matches the undamaged 80 m record, and the "10 m" echo time (0.402 μs) matches the damaged one. This suggests that these files were mislabelled in the public release. We report them but exclude them from the velocity estimate.

TABLE VII

ECHO TIMING ON THE WISE-CASING LABORATORY RECORDS (PULSE-TYPE TDR; NO ML APPLIED). DISTANCES USE THE VELOCITY FACTOR ESTIMATED FROM THE UNDAMAGED 80 M RECORD.

File	Documented setup	Doc. length (m)	Round trip (us)	Echo polarity	Implied vf	Distance at vf_est (m)	Consistent
TDR_20180315194711u.csv	TDR Shorted Termination Pre-Damage	80	0.836	negative	0.638	80.0	yes
TDR_20180315195720u.csv	TDR Shorted Termination with 120 Degree Cutaway	80	0.402	positive	1.328	38.5	n/a (damage echo)
TDR_20180404134819u.csv	TDR 20m Coaxial Cable Lab Experiment	20	0.842	negative	0.158	80.6	NO
TDR_20180404135441u.csv	TDR 10m Coaxial Cable Lab Experiment	10	0.402	negative	0.166	38.5	NO
TDR_20180404141951u.csv	TDR 5m Coaxial Cable Lab Experiment	5	0.137	negative	0.243	13.1	NO

TABLE VIII

LOCALISATION OF THE 40 M SHIELD CUTAWAY FROM THE ECHO-TIME RATIO.

Documented (m)	Echo-time ratio (m)	At nominal vf (m)	Error (m)	Echo polarity
40.0	38.5	39.8	-1.5	positive

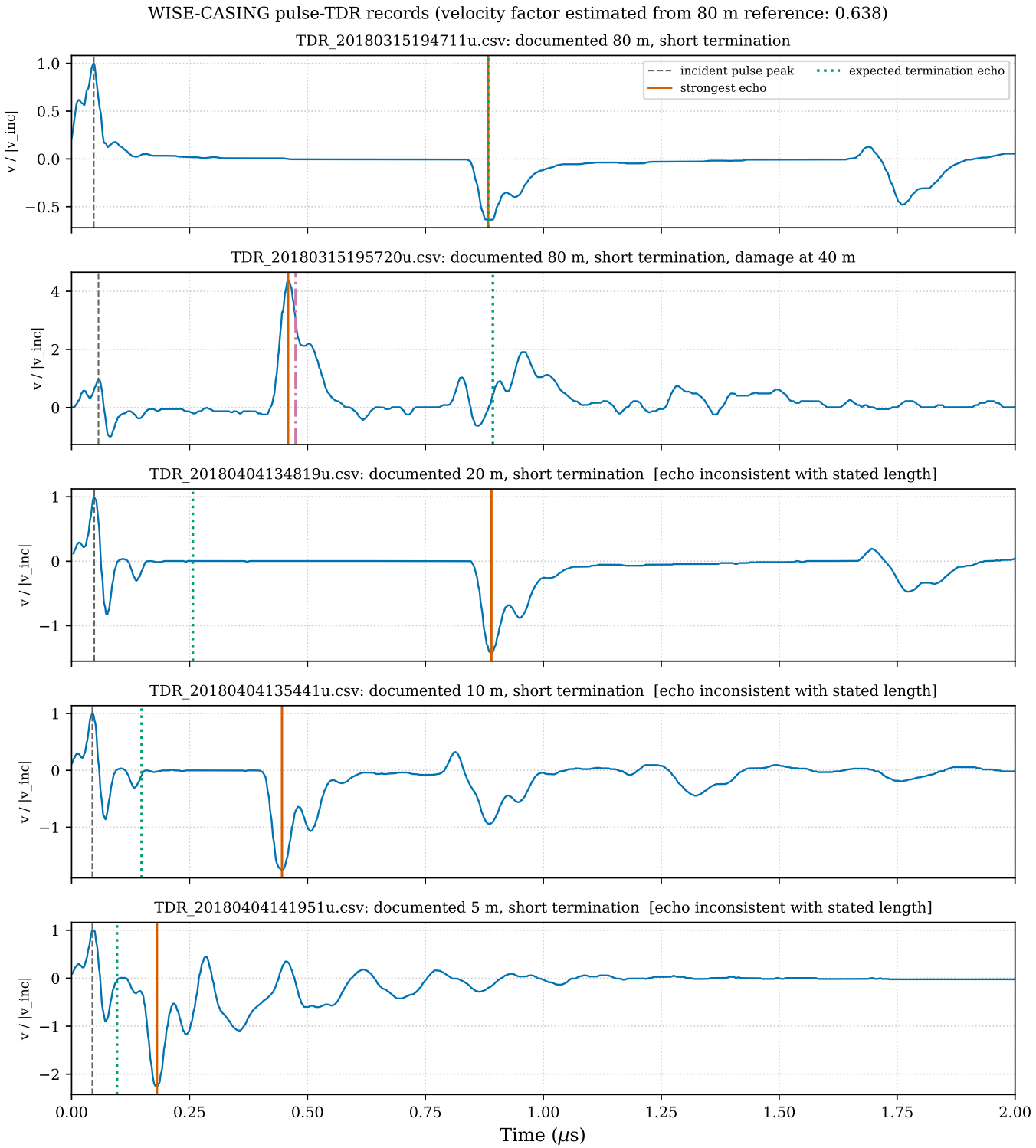


Fig. 14. WISE-CASING pulse-TDR records with the incident pulse peak, the strongest measured echo, and the echo time expected from the documented geometry at the velocity factor estimated from the undamaged 80 m record.

VII. Discussion, Limitations & Conclusion

A. Discussion of Findings

What machine learning adds over textbook TDR. The conventional detector is not a weak baseline where its assumptions hold. On hard faults and strong reflections it is as accurate as the learned models, and when it fires it locates the echo more precisely (median error 0.91 m). Its failures are systematic.

- Its detection threshold sits near $|\Gamma| \approx 0.1$–0.2 at 30 dB, so it misses a third of all faults.
- It cannot separate a weak fault from no fault.
- Its load identification breaks down on lossy lines, where the load edge is smeared.

The learned models are useful precisely in these regimes. They detect resistive faults about two to three times weaker, they classify loads above 93% at 30 dB, and they keep their advantage over the whole noise range and on every OOD shift. Both methods have physically meaningful limits: weak, short or capacitive discontinuities produce echoes that are hard to tell apart from noise, and no model trained on these data recovers them reliably.

Gradient boosting versus a 1-D ResNet. LightGBM is a strong, cheap and interpretable model: 94 features, about 16 min of CPU training per seed, and the lowest error on the load resistance. The ResNet is better on almost every other metric and is far more data-efficient. Trained on 3,000 traces, it already beats LightGBM trained on 60,000. It also degrades more gracefully below the training SNR range. Its price is higher seed-to-seed variance and a dependence on the training schedule (below). The SHAP analysis shows that LightGBM does not rely mainly on our hand-derived physics features (first-edge $\hat{\Gamma}$, Z/Z_0). It relies on the raw trace level just before the load echo and on edge timing. This suggests that much of our feature engineering was redundant with simply sampling the trace.

Input channels and multi-task learning. The ablations show that the Savitzky–Golay derivative and the coordinate channel are unnecessary. With a flattening head, the network can locate echoes in time from the voltage alone, and the first convolution learns its own derivative filter. Sharing one backbone across the three tasks is free for classification but costs about 1.2 m of location MAE compared with a location-only network. When location accuracy matters most, a dedicated location network, or a heavier location loss weight, is the better choice.

Location error. Two numbers summarise the location results: a 1.6 m median and a 6 m mean. Strong reflections ($|\Gamma| \geq 0.2$) are located to 1.0–1.7 m. The mean is dominated by weak faults, for which the regression head falls back towards the middle of the line (Fig. 15 shows such a case). A hybrid method is suggested by these results. The learned classifier would decide *whether* and *which* fault is present and give a coarse position. A conventional edge-timing step inside that window would then provide the sample-level precision that the rule-based detector already achieves.

Real hardware. The laboratory records confirm the propagation physics that the simulator and the location target rely on. Echo timing on a real 80 m coaxial cable gives a plausible velocity factor (0.638). The echo-time ratio places a real shield cutaway within 1.5 m of its documented position, with the polarity expected for a local impedance increase. We could not test the learned models on these data: the records are pulse-type TDR, and two of the five appear to be mislabelled.

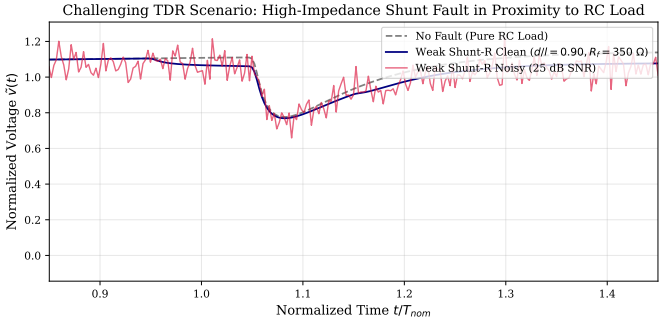


Fig. 15. A hard case: a 350 Ω shunt fault at $d/l = 0.90$ in front of an RC load. The fault echo ($|\Gamma| \approx 0.07$) is close to the noise level at 25 dB and overlaps the start of the load transient.

B. Limitations and Future Scope

- **Simulation-only evaluation of the ML models.** All learned-model results are on simulated traces. The only public laboratory data we found are pulse-type, so only the echo-timing physics was checked on hardware. A true sim-to-real test needs step-TDR measurements of lab coax with known faults, for example from a NanoVNA in time-domain mode.
- **Simplified line and instrument model.** The simulator covers one discontinuity and one termination on a single coaxial line, with skin effect and a constant loss tangent. It omits connectors, multiple or distributed faults, branched networks, temperature gradients, the instrument's own response (sampling jitter, baseline drift) and non-Gaussian interference. Each OOD subset changes only one factor at a time.
- **Near-load faults.** No method handles faults within 3 m of the load, and positions beyond $d/l = 0.95$ are outside the training range. Training on the full d/l range and adding an explicit "fault at the load" class would address the second point. The first is a resolution limit set by rise time and dispersion.
- **Early stopping on a noisy validation loss.** The validation loss fluctuates strongly from epoch to epoch (Fig. 16). As a result, the patience-8 rule ended 2 of the 12 ResNet runs early: seed 2 after 20 epochs, and the $(\tilde{v}, d\tilde{v}/dt)$ ablation after 15. The other runs saved their best checkpoint at or near epoch 40, so they were probably still improving. This inflates the reported seed variance, and it confounds the one ablation. Removing early stopping (keeping the best checkpoint) or training longer would likely improve the ResNet further.
- **Statistical scope.** The learning curves, ablations, noise sweep, OOD and error breakdowns use one seed (seed 0)

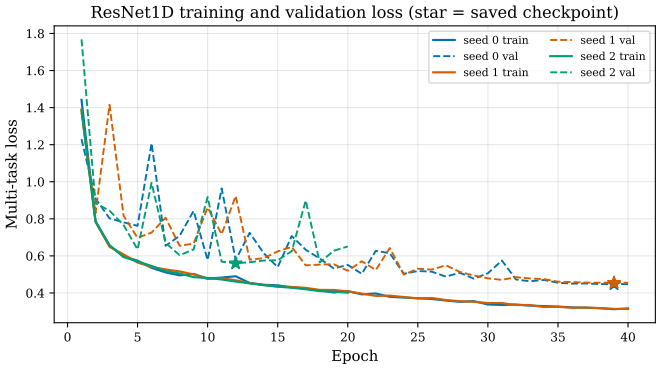


Fig. 16. ResNet training and validation loss for the three seeds. Stars mark the saved checkpoint; seed 2 was stopped by the patience rule at epoch 20.

and one fixed noise realisation per SNR level. Only the main comparison averages three seeds.

C. Conclusion

We built an end-to-end framework for identifying transmission-line faults and loads from a single step-TDR trace. It rests on a frequency-domain lossy-line simulator validated against bounce-diagram and transient theory. On it we compared a conventional rule-based detector, LightGBM on 94 engineered features, and a multi-task 1-D ResNet. On 10,000 held-out traces at 30 dB, the ResNet reached a fault macro-F_1 of 0.848 ± 0.007 (LightGBM 0.790, conventional 0.622), a location MAE of 6.06 ± 0.63 m (median 1.6 m; 1.0–1.7 m for $|\Gamma| \geq 0.2$) and a load accuracy of $96.3\% \pm 0.8\%$ (93.4% and 80.2%). The learned models move the detectable reflection from $|\Gamma| \approx 0.1$–0.2 down to about 0.05. They also stay ahead of the conventional method from 60 to 15 dB and under shifts in impedance, rise time, loss and cable length. All methods fail for faults within 3 m of the load. On real laboratory records, echo timing located a shield cutaway on an 80 m cable within 1.5 m. The main next step is a sim-to-real test of the learned models on measured step-TDR traces.

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